CMPE 255 · SHORT STORY ASSIGNMENT

Deep Learning Advancements in Anomaly Detection

A Comprehensive Survey — Huang et al., arXiv:2503.13195v1, March 2025

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A survey of deep learning for anomaly detection — what works, and why.

AT A GLANCE

180+
studies reviewed
mostly 2019–2024

3
families of methods
reconstruction · prediction · hybrid

99.4%
FastFlow AUC
MVTec AD, industrial defects

> 0.90
Chest CT AUROC
18 abnormalities

Huang et al. (2025)
IEEE Internet of Things Journal

Roadmap

Five parts — the one slide to remember is the map (slide 9)

1 **Foundations** Slides 3–5

2 **Data** Slides 6–7

3 **Applications** Slide 8

4 **Methods** Slides 9–14

5 **Evidence & outlook** Slides 15–18

Foundations

Slides 3–5

What an anomaly is, why it matters, and the paper at a glance.

Data

Slides 6–7

Data types, anomaly types, and the data problems that make this hard.

Applications

Slide 8

Where it is used: traffic, power grids, healthcare, finance, cyber.

Methods

Slides 9–14

Reconstruction, prediction and hybrids — in depth, with trade-offs.

Evidence & outlook

Slides 15–18

Results, ablations, open issues and the conclusion.

What is an anomaly?

A rare observation — with outsized consequences

1 Definition

An observation that deviates significantly from expected behaviour in a system. The survey's starting point, Section I.

2 Rare, but critical

Usually a tiny fraction of the data — often under 1% — yet frequently the most important part of it.

3 Examples

One fraud in a million transactions · one drifting sensor · one lesion on a scan · one attack in a log file.

4 Output = a decision

Investigate this, stop this, fix this now. Accuracy without action in time is a failure.

5 Why it is hard

Normal behaviour is broad and shifting, so 'different' is not the same as 'wrong' — context decides.

6 Key point

This is not classification for its own sake. The model exists to trigger a decision, in time.

Why it matters: the stakes

Missing it costs far more than a false alarm — until false alarms cost trust

1 Cost of missing

A missed anomaly can mean a factory shutdown, a fraud loss, a delayed diagnosis, or a cyberattack.

2 Cost of false alarms

A thousand false flags and the team stops trusting the system — then the real alerts get ignored too.

3 The metric tension

That trade-off runs through the whole survey, and it is why the choice of metric matters so much.

4 Why now

IoT put sensors everywhere: massive, high-dimensional, noisy, streaming data. Old methods broke on it.

5 Data got harder

Images, video, logs, vital signs — not small clean tables. Traditional methods assumed the opposite.

6 What changed

Deep learning learns representations directly from that data, instead of relying on hand-crafted features.

The paper at a glance

A survey, not a single model — and that is its strength

1 180+ studies

Mostly 2019–2024. Earlier surveys mostly stopped around 2020 and missed the newer models entirely.

2 Authors

Huang & Wang (York University) · Pei (HUST) · Wang & Niyato (NTU Singapore) · Alexanian (York).

3 Venues

IEEE, ACM, Springer, Elsevier journals · AAAI, CCS, ICCV conferences — top-tier sources throughout.

4 Gap it fills

New GAN variants — conditional, cycle-consistent, GAN + VAE — and hybrid models older surveys missed.

5 Contribution

Organisation: a taxonomy, structured comparison tables, and a full section on hybrid methods.

6 My focus

The gist, the architecture, the ablations and the metrics — not the maths in depth.

Data types & anomaly types

Name the data and the anomaly — before you pick a model

1 Text & audio

Text: incoherent word sequences, missing words in logs. Audio: distortion, frequency shifts, missing segments.

2 Image & video

Image: spatial defects, lesions. Video adds time: unusual movement, scene changes, motion inconsistency.

3 Time-series

Order matters. Point: one strange value. Sequence: a strange stretch. Group: a strange cluster together.

The survey's map — score: reconstruction error | prediction error | distance / density outside the boundary

RECONSTRUCTION	PREDICTION	HYBRID
AE · VAE · GAN · Diffusion	RNN · LSTM · Transformer · CNN · GNN	Clustering · Flows · SVDD

Train mostly on normal data → flag what deviates → judge by AUC, F1 and recall, not accuracy

Checklist before any code: data type, anomaly type, then model — in that order.

Why the data makes this hard

Real projects fail on data, not on architecture

1 Dimensionality

Curse of dimensionality: distances stop meaning much, so the model must learn a lower-dimensional representation itself.

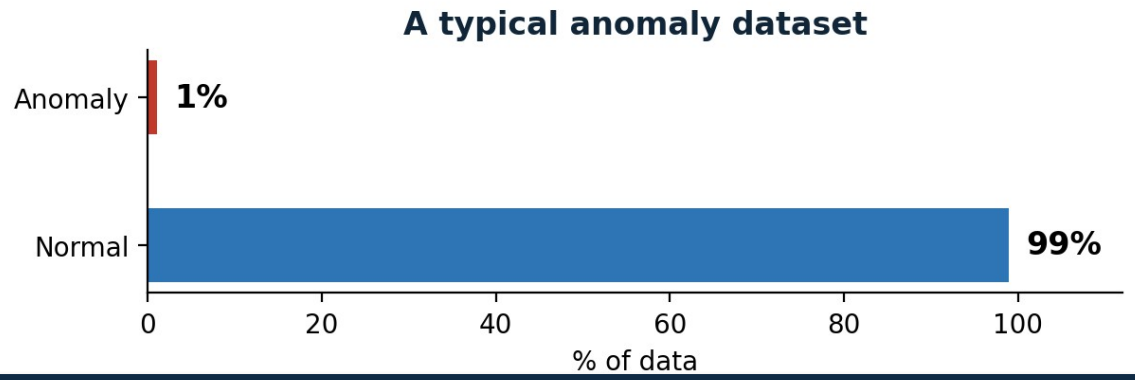
2 Class imbalance

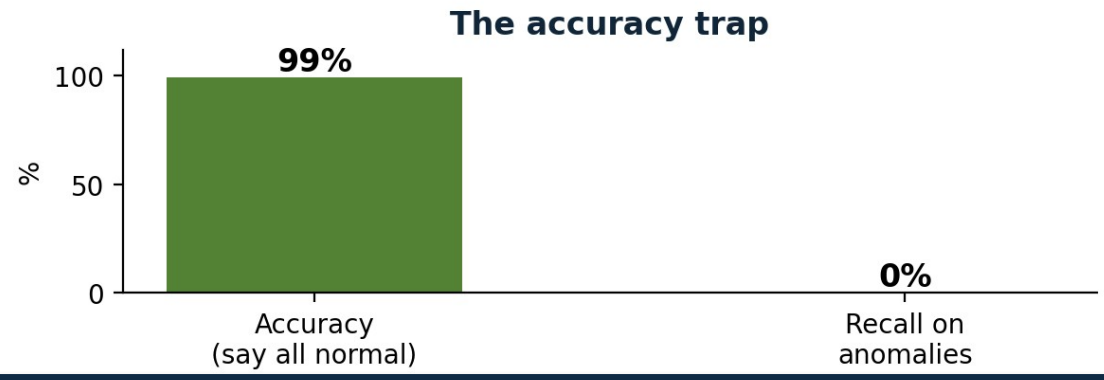
Anomalies are rare, so high accuracy can hide terrible recall. Judge by AUC, F1 and recall, not accuracy.

3 Noise & missing data

Noisy training data teaches an autoencoder to reconstruct the noise. Sensor data is also sparse and missing.

99% accuracy can mean 0% anomaly detection — judge by AUC, F1 and recall





If accuracy is your only metric on imbalanced data, your model can be 99% right and 100% useless.

Where it is used

Three worlds: time-series, non-temporal, visual

1 Time-series

Traffic monitoring with loop sensors · power grids — MENSA IDS detects cyberattacks · healthcare vitals, monitored continuously.

2 Non-temporal

Finance: fraud detection, credit scoring, trading activity. Cybersecurity: intrusion detection, unauthorised access.

3 Visual

Medical imaging: X-ray, brain, CT scans · video surveillance in public spaces · manufacturing defects on the line.

4 Proof point

Chest CT: AUROC > 0.90 for 18 abnormalities — but anomalies often look like normal tissue, so subtle cases are hardest.

5 Finance in detail

Bank fraud, insurance fraud, financial statement fraud and cryptocurrency fraud — each a different pattern.

6 Cybersecurity

Early work used autoencoders, deep belief networks and RNNs; newer work adds far more sophisticated models.

The big map: three families

Reconstruction · Prediction · Hybrid

1 Reconstruction

Learn what normal looks like, rebuild the input, and flag large reconstruction error. AE · VAE · GAN · diffusion.

2 Prediction

Learn the sequence, predict the next value, and flag large prediction error. RNN · LSTM · Transformer · CNN · GNN.

3 Hybrid

Deep features + a traditional, explainable boundary. Clustering · normalising flows · Deep SVDD.

The survey's map — score: reconstruction error | prediction error | distance / density outside the boundary



Train mostly on normal data → flag what deviates → judge by AUC, F1 and recall, not accuracy

Reconstruction: does it look normal? Prediction: does it follow? Hybrid: both, with a boundary you can explain.

Reconstruction 1: autoencoders & VAEs

Train on normal only — anomaly = high reconstruction error

1 Autoencoder

Compress → bottleneck → rebuild. Trained only on normal data, so an anomaly rebuilds badly and its error is high.

2 VAE

Latent space is a distribution (mean + variance). More principled score, and the model can generate new samples.

3 Strength

Works unsupervised, on high-dimensional data, with no labelled anomalies at all — the common real-world case.

Autoencoder — train on normal only: anomaly = high reconstruction error



Anomaly score = | input − rebuilt | • *small for normal data, large for anomalies* • *VAE: latent space is a distribution*

Train on normal only. An anomaly is whatever rebuilds badly.

Reconstruction 2: GANs & diffusion

More expressive — and harder to tame

1 GAN

Generator creates realistic normal data; discriminator judges real vs generated. Either side can signal an anomaly.

2 New variants

Conditional GANs, cycle-consistent GANs, GAN + VAE, GAN + self-supervised learning — missed by older surveys.

3 Diffusion

Learn to denoise step by step. An anomaly is a sample that does not denoise cleanly. Powerful, and the most expensive.

4 GAN + VAE hybrids

Combining the two covers each other's weaknesses — a recurring pattern in the survey's later papers.

5 Training cost

GANs can be unstable: generator and discriminator can fall out of balance, and tuning is fiddly.

6 Failure mode

Reconstruction error alone can be fooled — an anomaly that happens to reconstruct well is missed.

Prediction 1: RNNs, LSTMs & Transformers

Look at the past, predict the next value, measure the surprise

1 LSTM / GRU

Hidden state remembers what came before. Gates control what to remember and forget — still the workhorse.

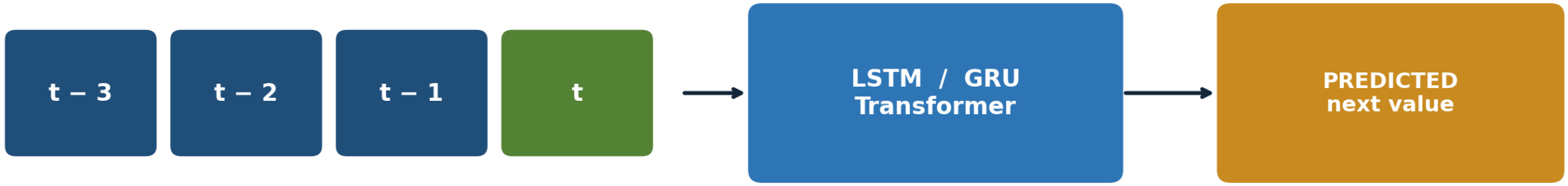
2 Transformer

Attention weighs time steps directly, instead of squeezing everything through a hidden state.

3 Results

F1 > 0.96 in one multimodal setting · up to +17% F1 from better inference in another study.

Prediction — learn the sequence: anomaly = large prediction error



Anomaly score = | actual next value − predicted next value | • *the threshold on this error decides the alarm*

A good forecast is not a good anomaly score. The threshold decides.

Prediction 2: CNNs & graph neural networks

An anomaly can be a broken relationship, not just a strange value

1 CNN

Local spatial / temporal patterns — a short sensor window, an image patch. Often combined with recurrent layers.

2 GNN

Sensors as a graph. A sensor fine alone but suddenly disagreeing with all its neighbours is a graph anomaly.

3 Strength

Temporal and relational modelling — the two things reconstruction alone cannot do.

4 Combinations

CNN + RNN is common: local detail from the CNN, long-term context from the recurrent layer.

5 Cost

Heavier to run in real time, and needs sequential or relational data to be useful at all.

6 When not to use

Static, independent data → use reconstruction. A graph model there is overkill, slower, and harder to debug.

Hybrids: old ideas, new power

Deep learning gives features — the traditional method gives a boundary

1 Clustering + deep

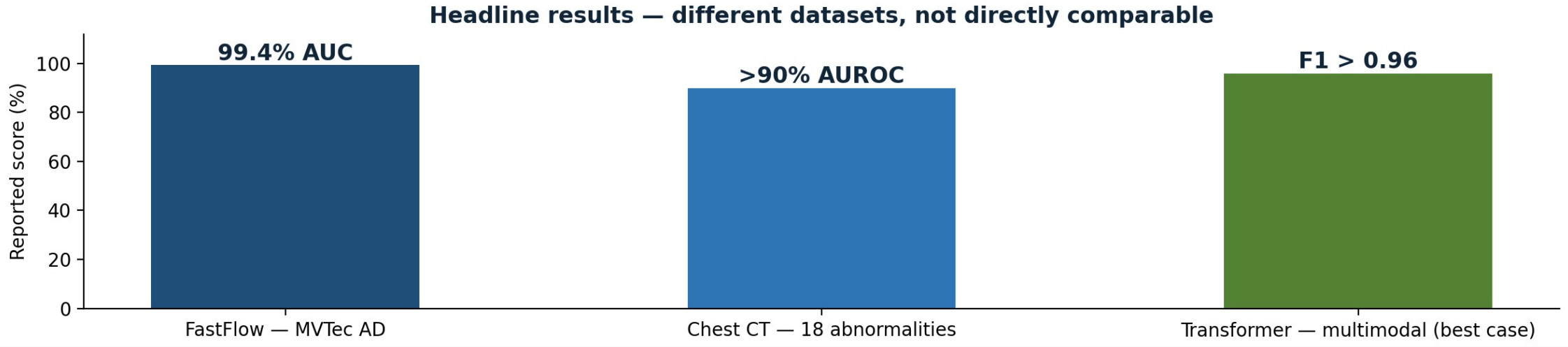
Extract features with a network, then cluster them. Keeps clustering's interpretability, gains deep representations.

2 Normalising flows

Invertible transforms estimate the exact density of normal data. Anomaly = low probability. FastFlow: 99.4% AUC, MVTec AD.

3 Deep SVDD

Smallest sphere around normal data in latent space; flag anything outside it. Beats AE methods on MNIST, CIFAR-10, GTSRB.



FastFlow at 99.4% AUC is the headline — because it is accurate and fast.

What actually works: results & metrics

Headline numbers — from different datasets, so read them carefully

1 FastFlow — 99.4% AUC

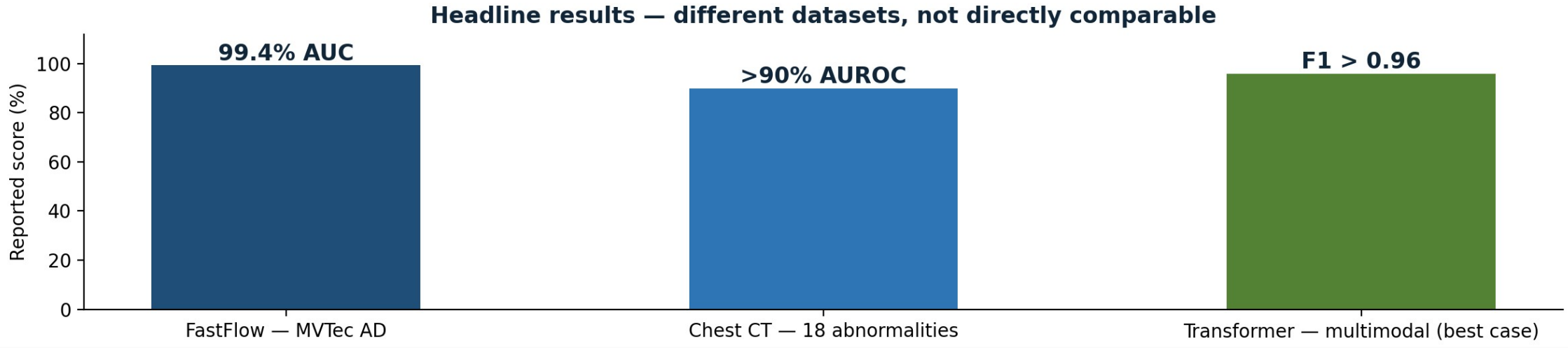
MVTec AD, industrial defects. Headline result — and it matters because it is both accurate and efficient.

2 Chest CT — AUROC > 0.90

18 abnormalities. Deep methods can work where the signal is subtle and close to normal tissue.

3 Transformer — F1 > 0.96

Multimodal setting, best case reported. Prediction wins where temporal dependencies are strong.



These numbers come from different datasets. Compare methods, not headlines.

Ablations & trade-offs — my reading

What the papers collectively say, when you read across them

1 Score > architecture

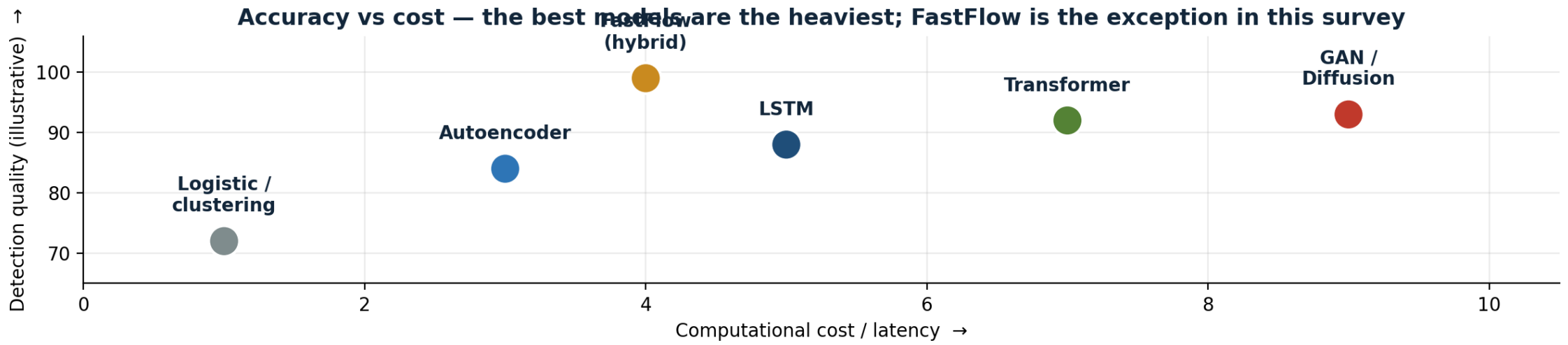
Swapping AE → VAE or LSTM → Transformer moves the score a few points. The wrong score / threshold breaks the system.

2 Data > model

Clean normal data, imbalance handling, noisy labels matter more than adding another layer.

3 Accuracy vs cost

Best models are the heaviest. A lighter model answering in milliseconds can be right for real-time use.



First ablation to run: fix the model, vary the threshold, plot precision vs recall.

Open issues & future work

The survey's honest ending — and one addition of my own

1 Data collection

Anomalies are rare, labelling is costly and needs experts, and synthetic data does not fully solve it.

2 Compute

AE / GAN / LSTM are heavy. Real-time needs lightweight models, GPU acceleration, edge / cloud offloading.

3 Explainability

Black box is not enough in healthcare, finance, safety-critical systems. Answers now: attention, LIME, SHAP.

4 Diverse anomalies

A model that catches point anomalies may miss sequence or group anomalies entirely.

5 Future directions

Self-supervised learning, more diverse synthetic data, adaptive frameworks, real-time explainability.

6 My addition: benchmarks

Same datasets, same splits, same metrics — until then, headline numbers will keep looking better than they are.

Conclusion

Thank you — questions welcome

1 Reconstruction

Does it look normal? Learn normal, rebuild the input, flag large reconstruction error.

2 Prediction

Does it follow? Learn the sequence, predict the next value, flag large prediction error.

3 Hybrids

Deep features + an explainable boundary — the strongest results in this survey, including 99.4% AUC.

4 Real-world success

Data quality, class imbalance, thresholds and explainability decide whether a model works in practice.

5 My take

Spend your time on the threshold and the evaluation, not on the architecture. That is where performance lives.

6 References

Paper, article, slides and full references are in the GitHub repo linked in the submission.